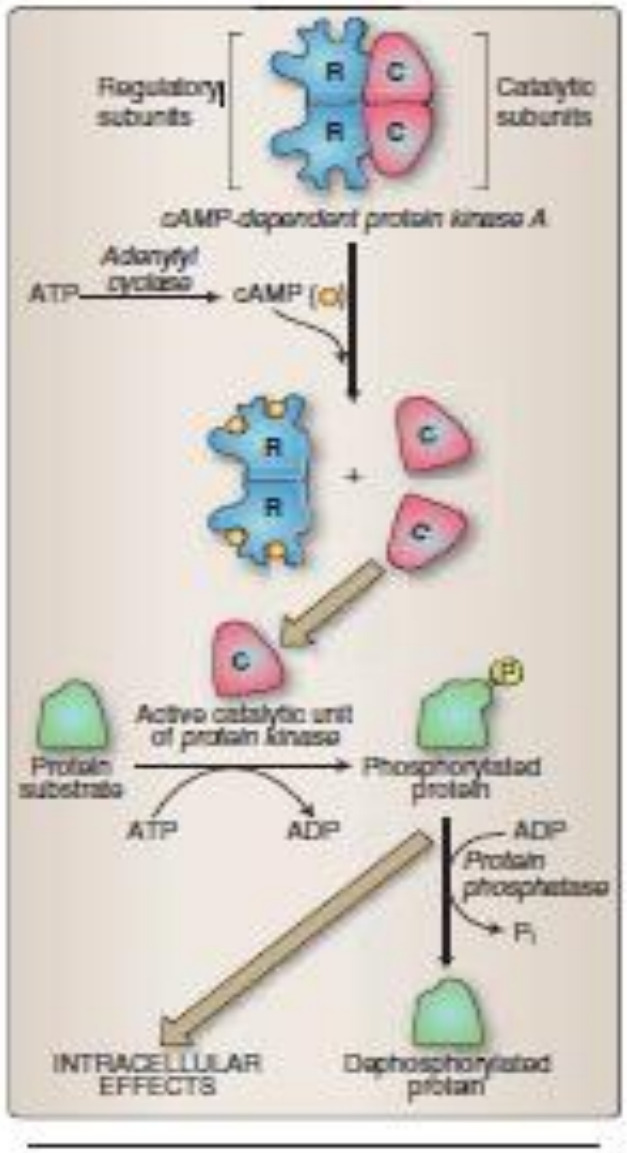
Cell communication and cell signaling

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Activation of PKA by cAMP



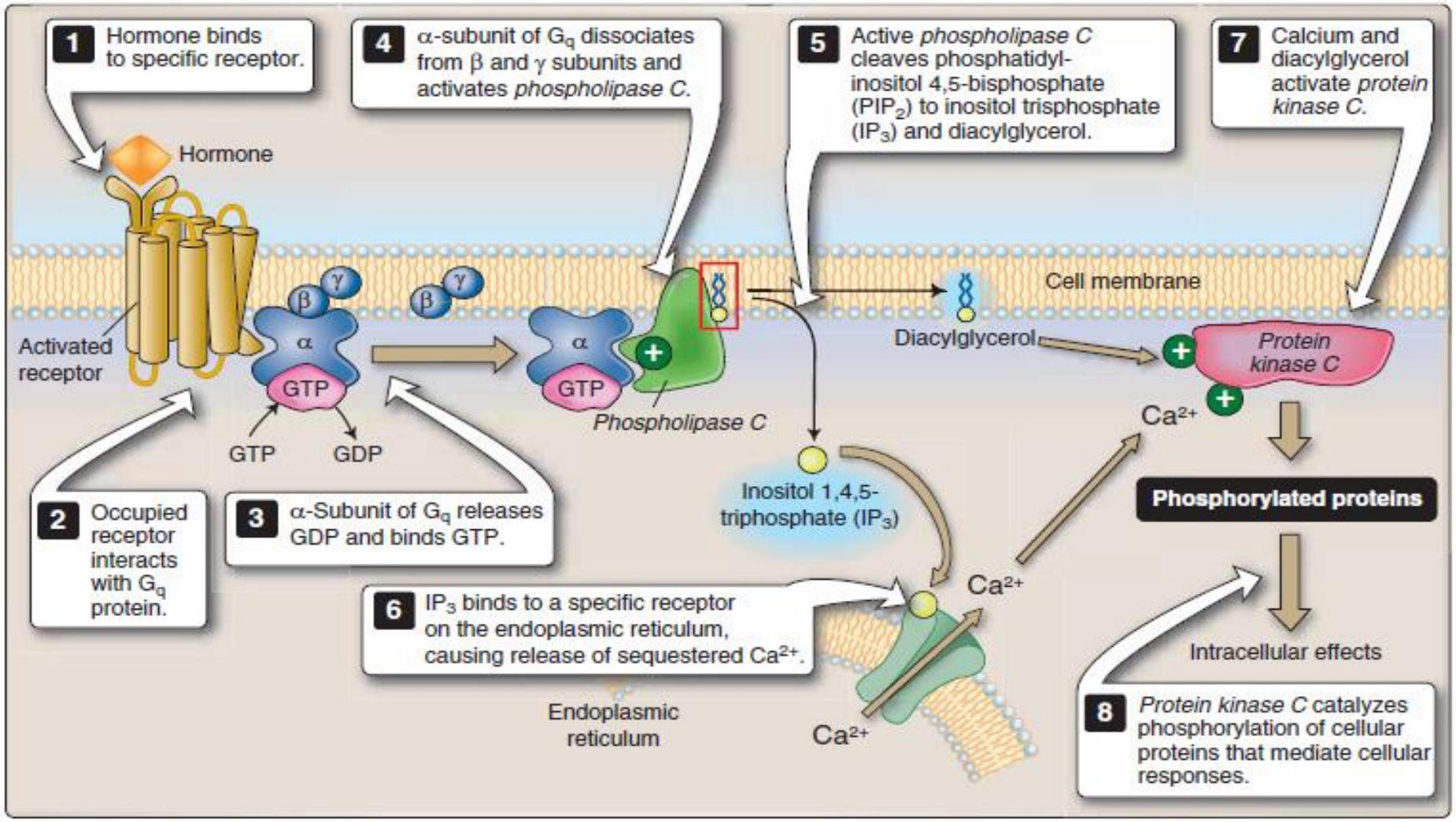
Toxins and Gα proteins that regulate adenylyl cyclase

Pertussis toxins alter Gα subunits and abnormal concentrations of cAMP in infected cells result. Cholera toxin is produced by ***Vibrio cholera* bacteria** that produce cholera toxin when they infect intestinal epithelial cells. This toxin modifies the Gαs subunit so that it cannot hydrolyze GTP and adenylyl cyclase remains active indefinitely. This devastating disease is caused by pertussis toxin produced by the infecting bacteria. It inhibits Gαi so that Gαi cannot inhibit adenylyl cyclase. Adenylyl cyclase remains active indefinitely, producing excess cAMP. **Symptoms:** Diarrhea and dehydration result from excessive outflow of water into the gut in response to excess cAMP. Cholera can be fatal without appropriate hydration therapy.

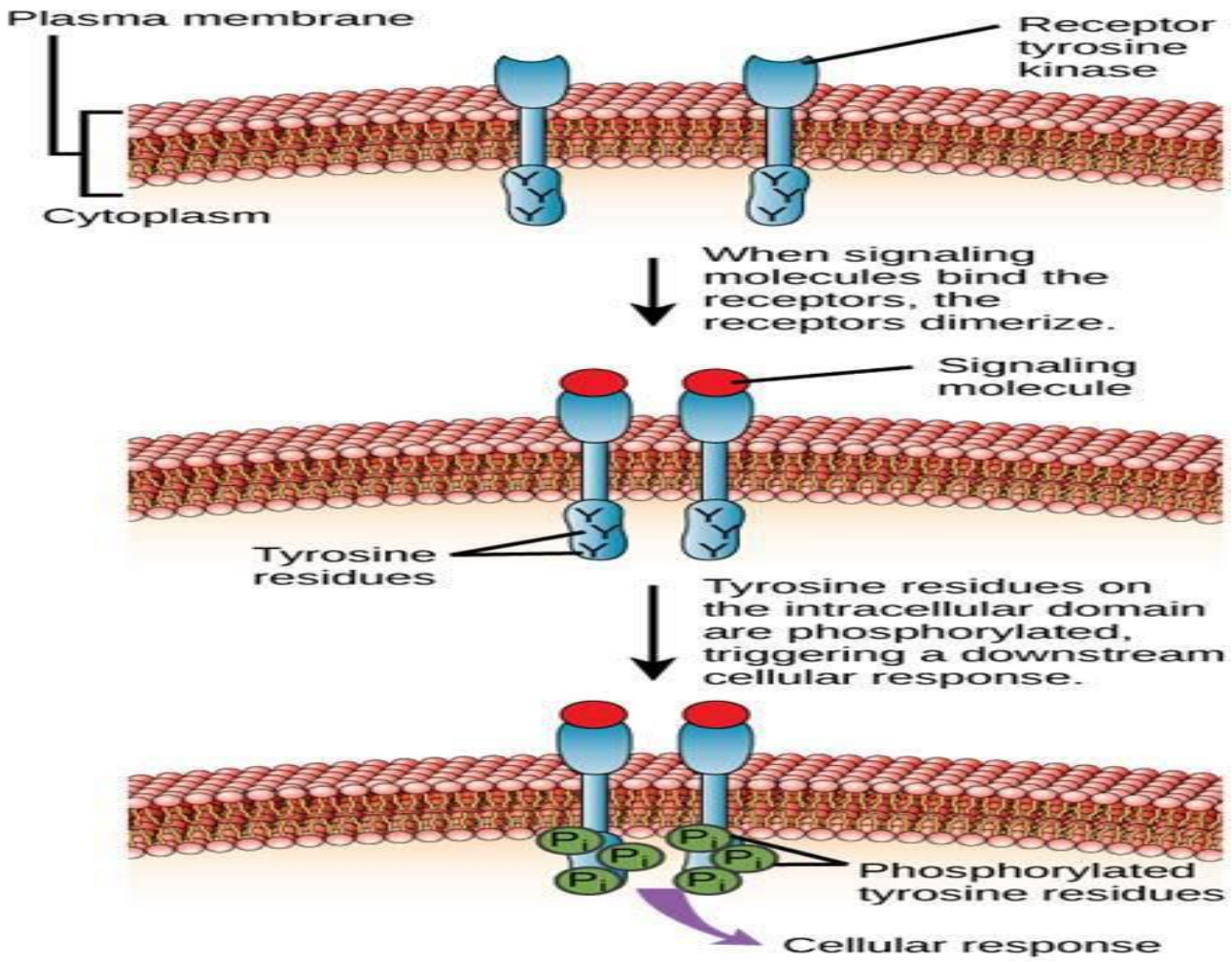
Bordetella pertussis is a bacterium that infects the respiratory tract and causes pertussis or whooping cough. Coughing can lead to vomiting and dehydration. **Antibiotics and hydration therapy** are used in treatment.

Vaccination now prevents many young children from dying from the effects of pertussis. However, it remains a major health threat. The World Health Organization reports that there were 39 million cases and 297,000 deaths attributed to pertussis in 2000. Ninety percent of all cases are reported in developing countries but the numbers of cases have been rising in the United States each year.

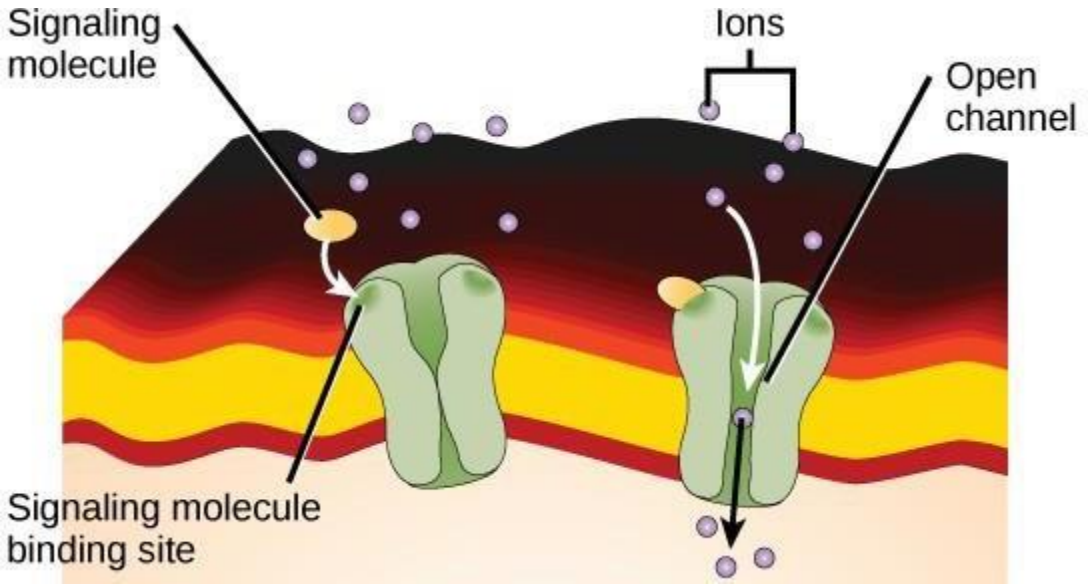
Generation of second messengers in response to Gq activation of phospholipase C



Tyrosine kinase receptor

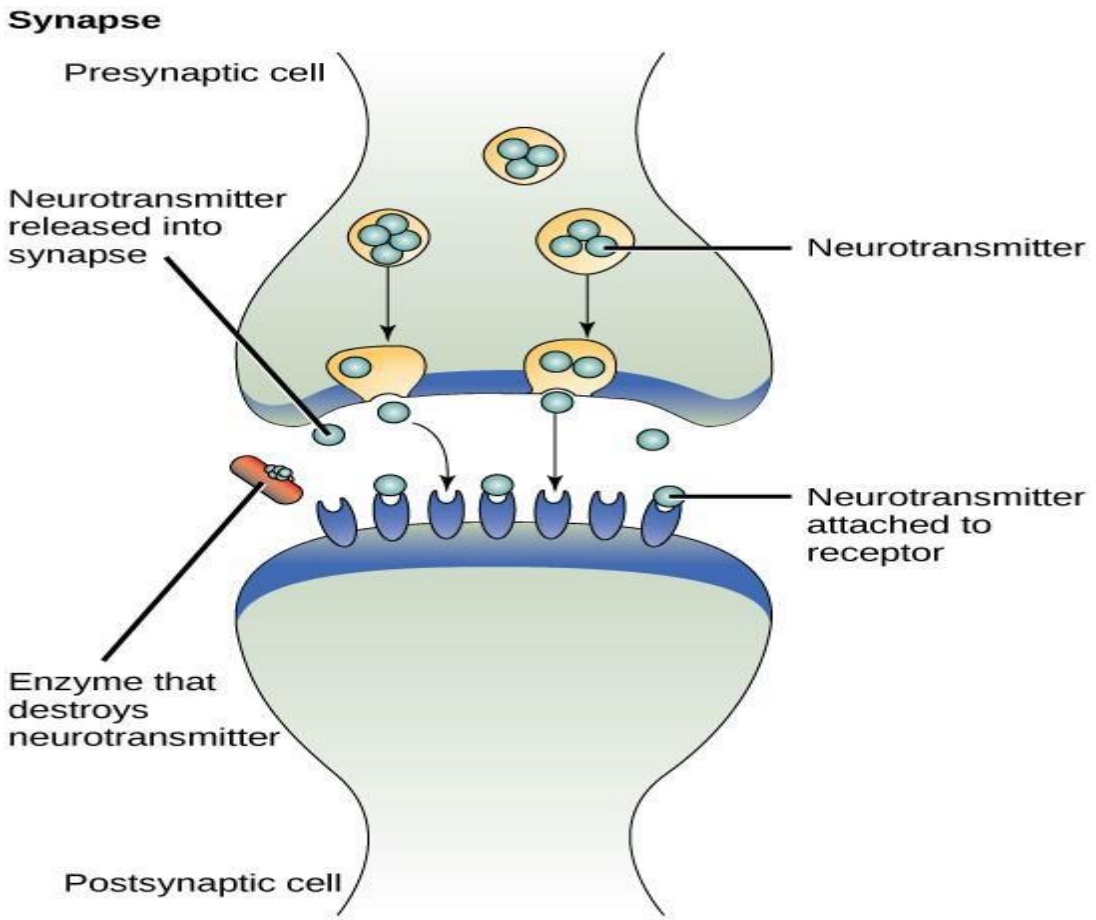


Ion channel signaling



Ion channel-linked receptors open and allow ions to enter a cell. An example of an ion channel-linked receptor is found on neurons. When neurotransmitters bind to these receptors, a conformational change allows sodium ions to flow across the cell membrane, causing a change in the membrane potential.

Neurotransmitter signaling (acetylcholoine)



Neurotransmitter signaling

One example of paracrine signaling is the transfer of signals between nerve cells.

The tiny space between nerve cells where signal transmission occurs is called a synapse. Signals are propagated along nerve cells by fast-moving electrical impulses. When these impulses reach the end of one nerve cell, chemical ligands called neurotransmitters are released into the synapse by the presynaptic cell (the cell emitting the signal).

The neurotransmitters diffuse across the synapse The small distance between nerve cells allows the signal to travel quickly, which enables an immediate response, such as, “take your hand off the stove!” When the neurotransmitter binds the receptor on the surface of the postsynaptic cell, the next electrical impulse is launched.

The neurotransmitters are degraded quickly or are reabsorbed by the presynaptic cell so that the recipient nerve cell can recover quickly and be prepared to respond rapidly to the next synaptic signal.

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Responses to the Signaling Pathway

- **Gene Expression**
- Some signal transduction pathways regulate the transcription of RNA. Others regulate the translation of proteins.
- **Increase in Cellular Metabolism**
- The activation of β-adrenergic receptors in muscle cells by adrenaline leads to an increase in cyclic AMP inside the cell. Adrenaline is a hormone produced by the adrenal gland that prepare the body for short-term emergencies.
- **Cell Growth**
- Cell signaling pathways also play a major role in cell division. Cells do not normally divide unless they are stimulated by signals from other cells. The ligands that promote cell growth are called **growth factors**. Most growth factors bind to cell-surface receptors that are linked to tyrosine kinases.
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Cell Death

- When a cell is damaged, superfluous, or potentially dangerous to an organism, a cell can initiate a mechanism to trigger programmed cell death, or apoptosis.
- Apoptosis allows a cell to die in a controlled manner that prevents the release of potentially damaging molecules from inside the cell. However, in some cases, such as a viral infection or uncontrolled cell division due to cancer, the cell's normal checks and balances fail. External signaling can also initiate apoptosis. For example, most normal animal cells have receptors that interact with the extracellular matrix, a network of glycoproteins that provides structural support for animal cells.
- The binding of cellular receptors to the extracellular matrix initiates a signaling cascade within the cell. However, if the cell moves away from the extracellular matrix, the signaling ceases, and the cell undergoes apoptosis. This system helps prevent cells from traveling through the body and proliferating out of control, as happens with tumor cells that metastasize.

